

Sales-analysis-dashboard

This is a Data Analytics project focused on sales performance.

Project Members

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❖ Project Description:

This project focuses on analyzing eCommerce sales data to uncover meaningful insights into sales performance, customer behavior, and product trends.

An interactive dashboard is built using **Power BI / Tableau** to visualize key metrics and support business decision-making.

❖ Business Problem:

- Lack of clear visibility into sales performance
 - Difficulty identifying top-performing products
 - Poor understanding of customer purchasing behavior
 - Inefficient regional sales strategy
 - No proper tracking of sales trends over time
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❖ Problem Statement:

The company aims to analyze sales data to improve revenue, identify growth opportunities, and make better business decisions using data-driven insights.

❖ Objectives:

- Analyze overall sales performance
- Identify top-performing products and categories
- Understand regional sales distribution

- Study customer spending behavior
 - Track sales trends over time
 - Build an interactive dashboard for business insights
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❖ **Tools & Technologies:**

- **SQL** → Data extraction & aggregation
- **Python / Excel** → Data cleaning
- **Power BI / Tableau** → Dashboard visualization

Libraries (if Python used):

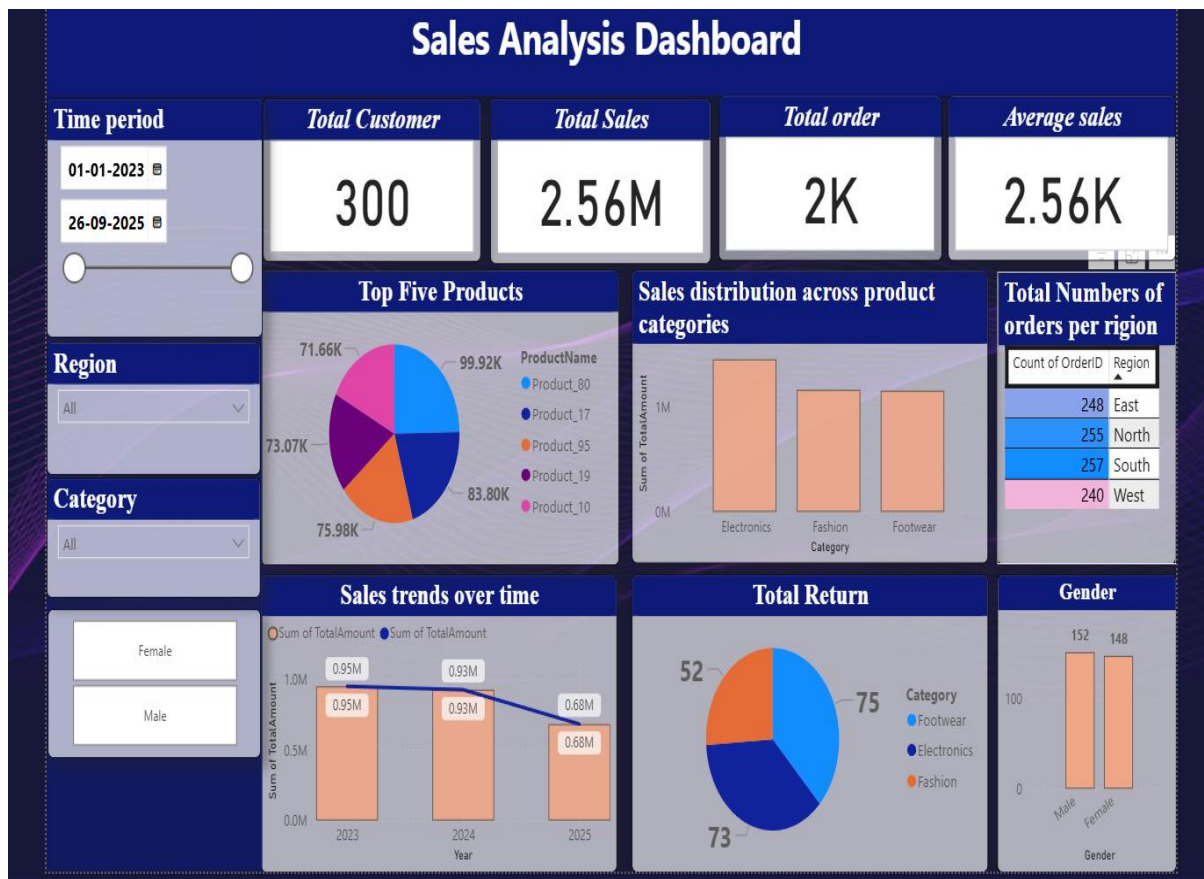
- Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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❖ **Methodology:**

- Data cleaning and preprocessing
 - Data aggregation using SQL
 - Exploratory Data Analysis (EDA)
 - Feature engineering
 - Dashboard creation
 - Insight generation
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❖ **Dashboard:**

Designed and developed an interactive dashboard to visualize sales insights.



❖ Dashboard Features:

- Total Sales KPI
- Total Orders KPI
- Average Sales KPI
- Sales Trend Over Time
- Category-wise Sales Analysis
- Region-wise Sales Comparison
- Top-performing Products
- Customer Segmentation (based on spending)
- Interactive filters (Date, Region, Category)
- Clean and user-friendly design

❖ **Exploratory Data Analysis (EDA):**

Performed to understand sales patterns, customer behavior, and product performance.

❖ **Key Analysis:**

- Analyzed sales trends over time
 - Compared sales across different regions
 - Studied product category performance
 - Identified top-selling products
 - Performed customer segmentation based on total spending
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❖ **Insights:**

- A small number of products generate most of the revenue
 - Certain regions contribute significantly higher sales
 - Sales show seasonal trends
 - High-value customers contribute major revenue
 - Some categories underperform and need improvement
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❖ **Strategic Recommendations:**

- **Focus on top-performing products**
Increase marketing and inventory for high-selling products
- **Improve low-performing categories**
Offer discounts or redesign product strategy
- **Target high-value customers**
Provide personalized offers and loyalty programs
- **Optimize regional strategy**
Invest more in high-performing regions

- **Use seasonal trends for planning**
Adjust pricing and stock based on demand patterns
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❖ **Conclusion:**

This project demonstrates how sales data can be transformed into meaningful insights using analytics and visualization tools. The dashboard helps businesses make informed decisions and improve overall performance.